

Unit 4 :: Data Preparation and Management

❧ SEC-03-PHY-91-R-03 ❧

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Syllabus

Unit 4: Data preparation and management

Basic pre-processing of data: filtering, sorting, normalization and handling missing values; managing multisheet workbooks and structured datasets, importing data from diverse file formats (CSV, ASCII, DAT), and ensuring interoperability between different spreadsheet and analysis platforms.

1 Importing data from diverse file formats

Suppose a physics experiment records a variable which is of interest to us, such as temperature, pressure, voltage, count rate etc. These variables are recorded (or sampled) at a sequence of time instants. Now this raw record of data produced by an instrument is almost certainly unfit for direct analysis because

- a) It may contain unreliable readings.
- b) It may be out of chronological order.
- c) It may mix standard and non-standard units.
- d) It may contain gaps.

Then we say that such a dataset must be *pre-processed* before it can be visualised or fitted to a theoretical curve.

For the purpose of pre-processing, the dataset must be first stored and archived in the form of a file. Three file formats are widely physics laboratory exports:

1. Comma-separated values (CSV files)
2. Instrument-specific data (DAT files)
3. Microsoft Excel spreadsheets (XLSX files)

1.1 ASCII files

ASCII, which is an acronym for *American Standard Code for Information Interchange*, is a character encoding standard. Originally, it was used for representing a particular set of 95 printable characters and 33 control characters, thereby giving a total of 128 code points, using 7 bits. Later, an 8-bit encoding extended the ASCII set to a total of 256 characters.

An ASCII file is one that contains data made up of ASCII characters, i.e. it is essentially raw text. Each byte in the file contains one character that conforms to the standard ASCII code.

Source codes of computer programs, batch files, macros and scripts are straight text and stored as ASCII files. HTML and XML files are also ASCII files. Text editors such as Notepad create ASCII files as their native file format.

CSV and DAT files are two of the most commonly used ASCII file formats used for storing scientific data for subsequent analysis.

1.1.1 CSV files

A comma-separated values (CSV) file is a plain text data format for storing tabular data where the fields (values) of a record are separated by a comma and each record is a line (i.e. newline separated). For example, the following are the contents of a CSV file:

```

1      id, name, email
2      1, John, john.doe@example.com
3      2, Jane, janey72@test.org

```

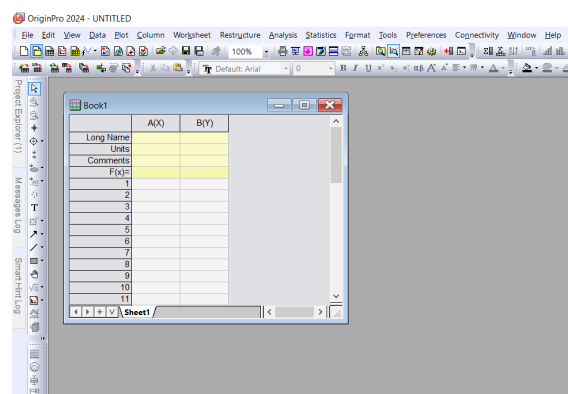
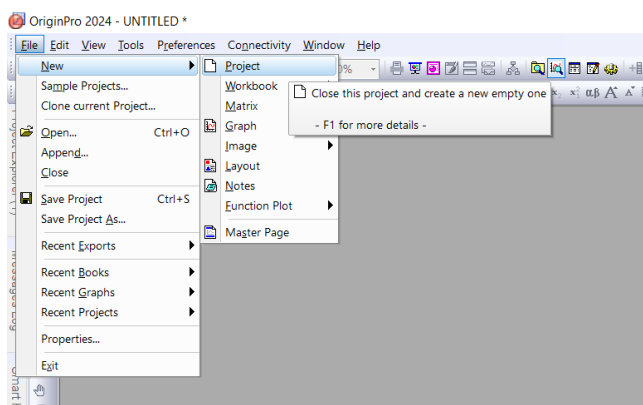
In software that generally deals with tabular data such as a database or a spreadsheet, the contents of the above CSV file is parsed in the form of a table as follows:

id	name	email
1	John	john.doe@example.com
2	Jane	janey72@test.org

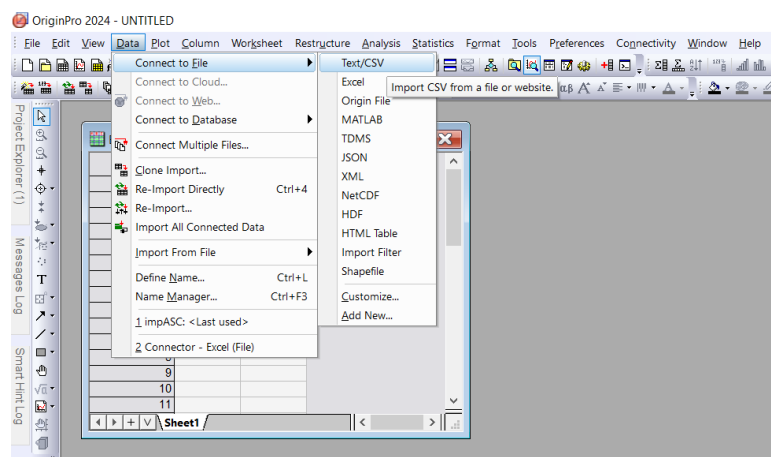
The first line is used as the table header and the remaining lines are used as the rows of the table.

Importing data from a CSV file in Origin:

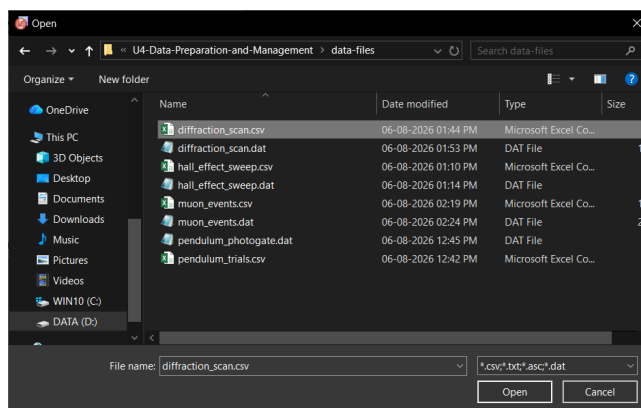
- I. In the main menu of Origin, go to File > New > Project. This would create a new Origin project with an empty worksheet.



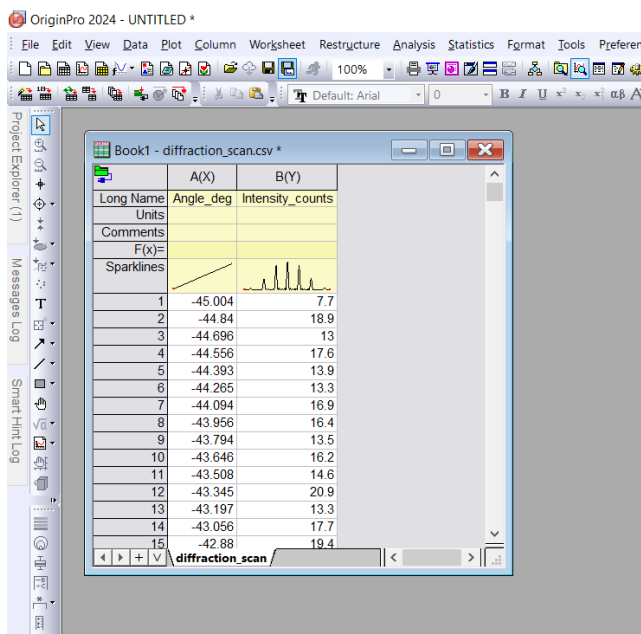
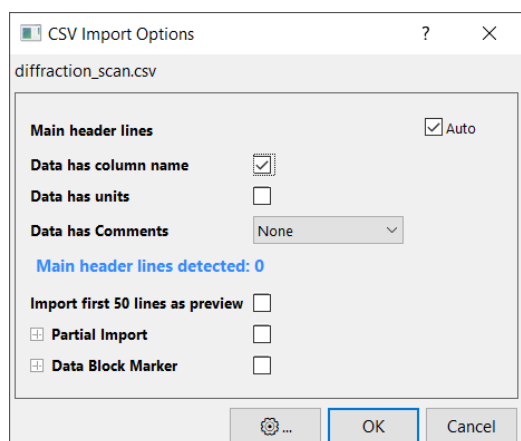
- II. To fill in the worksheet with values from the CSV file, we go to Data > Connect to file > Text/CSV.



III. In the file open dialog, we navigate to the folder containing the CSV file. Selecting the file of interest, we click Open.



IV. Another dialog appears where we can provide the options for importing data from the CSV file. Keeping everything unchanged, we click OK. We find that the data from the CSV file has been imported into the worksheet and is ready for further analysis.



1.1.2 DAT files

Computer programs create DAT files to store specific information or data. The information in the file is only relevant to the program that created it, though almost any program can create DAT files. DAT might look like an acronym, but the extension name is actually short for 'data.'

A DAT file may begin with header lines that begin with #. Any line that begins with a # is treated like a comment and is subsequently ignored. So we can make use of these header lines to insert specific information in the DAT file. However, it is possible to have DAT files without such header lines.

The actual data in a DAT file may be entered with each record in its own row and separated by whitespaces, such as space or tab. For example, the following are the contents of a DAT file:

```

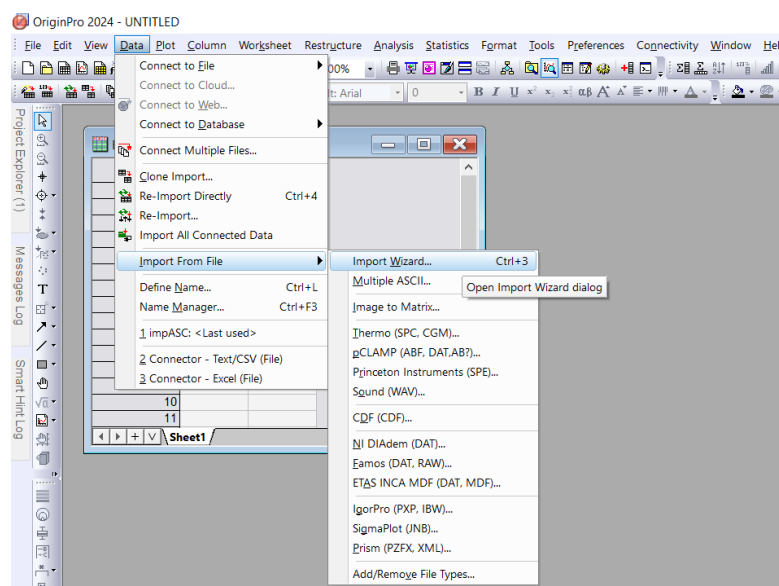
1      # Diffraction Grating Intensity Scan
2      # Wavelength_nm: 632.8
3      # Grating_spacing_nm: 3000.0
4      # Slit_width_nm: 800.0
5      # Date: 2026-08-01
6      Angle           Intensity
7      degree          counts
8      -45.000         10.5
9      -44.850          9.6
10     -44.700         16.3
11     -44.550         11.5
12     -44.400         15.2

```

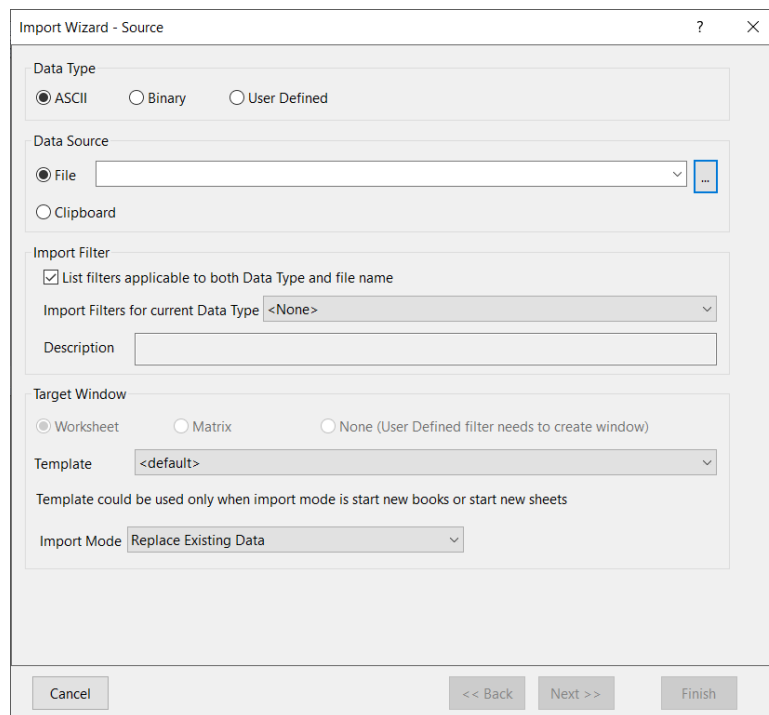
In the above DAT file, lines 1 to 5 are indicated as headers. Lines 6 and 7 are provide the column labels and units respectively. The subsequent lines contain the data, separated by a tab space.

Importing data from a DAT file in Origin:

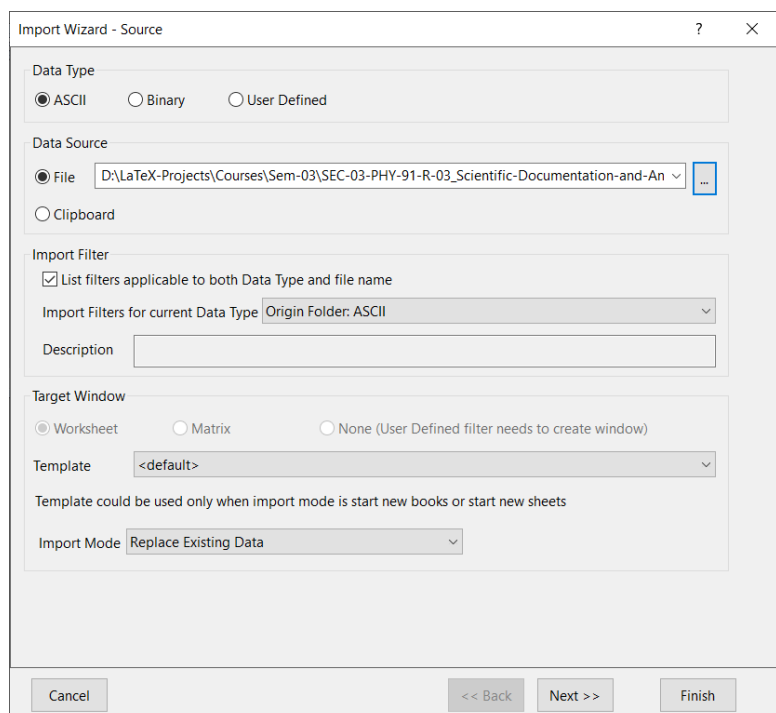
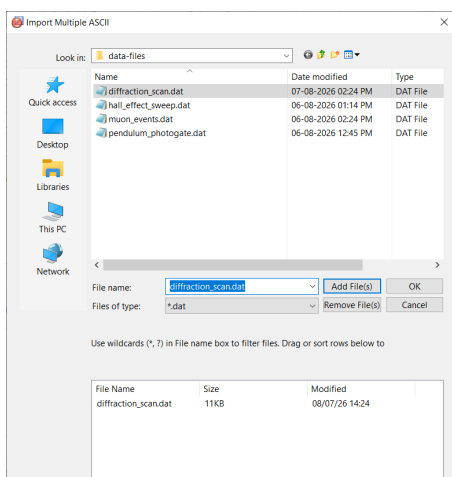
- I. Create a new project in Origin. Then to import data from a DAT file, go to Data> Import From File> Import Wizard.



- II. In the Import Wizard dialog box, we ensure that under Data Source, the File option is checked and then we click on the button to the right of the filename textbox.



- III. After navigating to the folder that contains the DAT file of interest, we select the file (or multiple files). Then we click Add File(s) and finally OK. We would be taken back to Import Wizard dialog where we can confirm that the full path of the selected DAT file is displayed in the filename textbox.



IV. Clicking Next on the Import Wizard dialog takes us to the next window where we set the options for data import.

We uncheck the Auto determine header lines checkbox. Then we enable the By Leading Character radio button and enter # in the textbox alongside. This is because in our DAT files, the header lines are indicated by the character # at the beginning.

We enter the input Number of subheader lines = 2 because two lines have special meanings as containing the long names of the columns and their units. We indicate these by the inputs Long Names = 1 and Units = 2.

A preview of the DAT file at the bottom of the dialog page confirms that our options are correct by displaying the header lines in black, the long names in blue, the units in green and the data in grey.

Import Wizard - Header Lines

☐ Auto determine header lines Main Header ☐ Fixed Number 5 ☒ By Leading Character #

Number of subheader lines 2 ☐ Line number start from bottom

Column Label Assignment from Subheader Lines

Long Names 1 Comments <None> to 0

Units 2 System Parameters <None> to 0

User Parameters <None> to 0

☐ Extract Long Names and Units from Same Line Characters to skip on each line 0

Preview Font System Preview Lines 50

Prefix: S=Short Name, L=Long Name, U=Units, P=Parameters, C=Comment, MH=Main Header, SH=Subheader

```

001 001MH # Diffraction Grating Intensity Scan
002 002MH # Wavelength_nm: 632.8
003 003MH # Grating_spacing_nm: 3000.0
004 004MH # Slit_width_nm: 800.0
005 005MH # Date: 2026-08-01
006 001SH L Angle Intensity
007 002SH U degree counts
008 -45.000 10.5
009 -44.850 9.6
010 -44.700 16.3
011 -44.550 11.5
012 -44.400 15.2
013 -44.250 12.2
014 -44.100 11.4
015 -43.950 16.3
016 -43.800 8.1

```

Cancel << Back Next >> Finish

V. We click on Next for the remaining dialog pages and finally click on Finish.

Import Wizard - Save Filters

Import Wizard settings can be saved to a filter file for re-use. Filter files can be selected on the first page of this wizard. Once saved, import filters can also be used to automatically determine import settings when dragging and dropping data files into Origin and when opening data files with the File/Open menu item.

☒ Save filter

☐ In the data file folder D:\LaTeX-Projects\Courses\Sem-03\SEC-03-PHY-91-R-03_Scientific-Documenta

☒ In the User Files folder C:\Users\Asus\Documents\OriginLab\User Files\Filters\

☐ In the Window

☐ Show Filter in File/Open List

Filter Description

Filter file name (.OIF extension will be appended) diffraction_scan

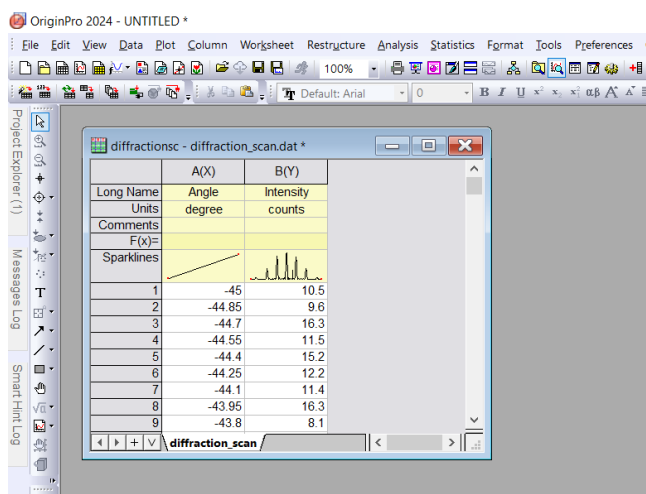
Applicable File diffraction_scan*.dat

Use wild cards such as *.txt, s???*.dat etc. Specify multiple names by ;, such as a*.dat; b*.dat

☐ Specify advanced filter options

Cancel << Back Next >> Finish

VI. When the Import Wizard closes, we find that the data from the DAT file has been correctly imported into the Origin workbook and it is now ready for analysis.



1.2 Interoperability with spreadsheets

2 Insert title

Insert content.

3 Insert title

Insert content.

4 Insert title

Insert content.

5 **Insert title**

Insert content.

6 Insert title

Insert content.